

$$a. \sin^4(xy^2) - 2x^2y - 10 = 0$$

$$\frac{dx}{dy} \Rightarrow 4\sin^3(xy^2) \cos(xy^2) (y^2 + xy \frac{dy}{dx}) - 2(2xy + x^2 \frac{dy}{dx}) - 0 = 0$$

$$= (4\sin^3(xy^2) \cos(xy^2) y^2 - 2x^2) \frac{dy}{dx} = 4xy - 4\sin^3(xy^2) \cos(xy^2) x$$

$$= \frac{dy}{dx} = \frac{4x^2 - 4y^2 \sin^3(xy^2) \cos(xy^2)}{4xy \sin^3(xy^2) \cos(xy^2) - 2x^2}$$

$$= \frac{2xy - 2y^2 \sin^3(xy^2) \cos(xy^2)}{2xy \sin^3(xy^2) \cos(xy^2) - x^2}$$

$$b. x^4 \tan(xy) + xy^3 + 5 = 0$$

$$= 4x^3 \tan(xy) + x^4 \sec^2(xy) (y + x \frac{dy}{dx}) + (y^3 + 3xy^2 \frac{dy}{dx}) + 0 = 0$$

$$= (x^4 \sec^2(xy) x + 3xy^2) \frac{dy}{dx} = -4x^3 \tan(xy) - x^4 \sec^2(xy) y - y^3$$

$$\frac{dy}{dx} = \frac{-4x^3 \tan(xy) - x^4 y \sec^2(xy) - y^3}{x^5 \sec^2(xy) + 3xy^2}$$

$$2. x^2y^2 - 6xy + 5 = 0$$

$$y = 1 \Rightarrow x^2 - 6x + 5 = 0$$

$$(x-5)(x-1)$$

$$x = 5 = x = 1$$

$$(1,1) \Rightarrow \frac{d}{dx} = (2xy^2 + x^2 2y \frac{dy}{dx}) - 6(y + x \frac{dy}{dx}) + 0 = 0$$

$$(x^2 2y - 6x) \frac{dy}{dx} = 6y - 2xy^2$$

$$m = \frac{6y - 2xy^2}{2x^2y - 6x} = \frac{6-2}{2-6} = \frac{4}{-4} = -1$$

$$\text{Pers} \Rightarrow y - 1 = -1(x - 1)$$

$$y = -x + 2 //$$

$$(5,1) \Rightarrow m = \frac{6y - 2xy^2}{2x^2y - 6x} = \frac{6-2}{2 \cdot 25 - 30} = \frac{4}{20} = \frac{1}{5}$$

~~$$x^2 y^2 - 6xy + 5 = 0$$

$$m = 1 \Rightarrow x^2 - 6x + 5 = 0$$

$$(x-5)(x-1)$$~~

$$(5, 1) \Rightarrow m = \frac{6y - 2x^2 y^2}{2x^2 y - 6x} = \frac{6 - 2 \cdot 5}{2 \cdot 25 - 6 \cdot 5} = \frac{-4}{20} = -\frac{1}{5}$$

$$\text{Pers} = y - 1 = -\frac{1}{5}(x - 5)$$

$$SV = -x + 10$$

$$3. f(x) = \frac{x^2 - 3x + 1}{x - 3}$$

$$\text{asimtot tegak} = x = 3$$

$$\text{asimtot datar} = \infty \Rightarrow \text{tidak ada}$$

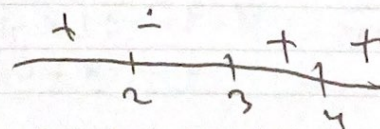
$$\begin{aligned} \text{asimtot miring} &= x - 3 \frac{x^2 - 3x + 1}{x^2 - 3x} \\ &= y = x \end{aligned}$$

Pemotongan

$$f(x) = \frac{x^2 - 3x + 1}{x - 3}$$

$$\begin{aligned} f'(x) &= \frac{(2x - 3)(x - 3) - (x^2 - 3x + 1)}{(x - 3)^2} \quad (1) \\ &= \frac{2x^2 - 9x + 9 - x^2 + 3x - 1}{x^2 - 6x + 9} \\ &= \frac{x^2 - 6x + 8}{x^2 - 6x + 9} = 0 \\ &= \frac{(x - 2)(x - 4)}{(x - 3)^2} = 0, x \neq 3 \end{aligned}$$

$$x = 2 \vee x = 4 \vee x = 3$$



$f(x)$ turun pada $(2, 3)$,
 $f(x)$ naik pada $(3, 4)$ & $(4, \infty)$